Carlos Jurado

Academic Website — carlosjurado@utexas.edu

Education

PhD, Astronomy; The University of Texas at Austin 2024 -
Advisor: Keith Hawkins

MA, Astronomy; The University of Texas at Austin 2026
Thesis: Dynamical Origins of Azimuthal Metallicity Variations in the Galactic Disk
Advisor: Keith Hawkins

BS, Astrophysics; University of California, Los Angeles (UCLA) 2024

Research

Graduate Student Researcher at UT Austin – Austin, TX Sept. 2024 – Present
Advisor: Keith Hawkins

Student Researcher at University of Notre Dame – Notre Dame, IN May 2023 – Aug. 2024
Advisor: Lauren Weiss

Combined radial velocity measurements of Kepler-34 and Kepler-35 from several spectrographs to place a strong mass upper limit on additional stellar and substellar companions in the two systems.

Student Researcher at UCLA – Los Angeles, CA May 2022 – June 2024
Advisor: Smadar Naoz

Utilized supercomputers to create population synthesis models and explore the spatial distribution of stellar remnants at the Galactic Center.

Publications

[1] **Jurado, C.**; Hawkins, K.; Hunt J. A. S.; Hackshaw, Z.; Filion C.; Frankel, N., Carr, C., "Dynamical Origins of Azimuthal Metallicity Variations in the Galactic Disk: Insights from kinematic ridges with Gaia", ApJ, 10.3847/1538-4357/ae346c

[2] **Jurado, C.**; Weiss, L. M.; Daclison L.; Tofflemire, B.M.; Orosz J. A.; Welsh, W. F., "Upper Limits on Stellar Companions to the Kepler-34 and Kepler-35 Systems", AJ, 10.3847/1538-3881/ada5f5

[3] **Jurado, C.**; Naoz, S.; Lam, C. Y.; Hoang, B.-M., "Natal Kicks from the Galactic Center and Implications on their Environment and the Roman Space Telescope," ApJ, doi.org/10.3847/1538-4357/ad55ee

Conference Presentations

Division on Dynamical Astronomy Annual Meeting, Chicago, IL June 2026
Contributing Talk: "The Dynamical Origins of Azimuthal Metallicity Variations"

APS Far West Section, San Diego, CA — Oct. 2023
Contributing Talk: "The implications of Exploding Stars at the Galactic Center"

Seminars & Informal Talks

Stars, Planets, and ISM Seminar, The University of Texas at Austin — April 8 2026
Talk: "Exploring the Chemo-Kinematic Structure of the Milky Way Disk"

Smadar Naoz Research Group Meeting, UCLA — Jan. 8 2026
Talk: "Dynamical Origins Of Azimuthal Metallicity Variations"

Stars, Planets, and ISM Seminar, The University of Texas at Austin — Jan. 29 2025
Talk: "The Effects of Natal Kicks in the Galactic Center"

AstroBash, UCLA Department of Physics and Astronomy — Oct. 18 2023

AstroBash, UCLA Department of Physics and Astronomy — Nov. 7 2022

Teaching

Graduate Teaching Assistant, Stellar Astronomy (AST 352K) — Aug. 2026 – Dec. 2026

Graduate Teaching Assistant, Search For Extraterrestrial Life (AST 309L) — Aug. 2025 – Dec. 2025

Guest Lecture, Introductory Astronomy (AST 307) — April 1, 2025
Lecture: "The Structure And Components Of The Milky Way Galaxy"

Graduate Teaching Assistant, Introductory Astronomy (AST 301) — Jan. 2025 – May 2025

Undergraduate Course Reader, Introductory Physics (Physics 5A) — Oct. 2023 – Dec. 2023

Undergraduate Course Reader, Introductory Mechanics (Physics 1A) — April 2024 – June 2024

Undergraduate Learning Assistant, Introductory Mechanics (Physics 1A) — Jan. 2023 – Mar. 2023

Awarded Telescope Time

McDonald Observatory — Feb., July & Nov. 2026
PI: Carlos Jurado (15 nights)

Spectroscopically observe 10 stellar triples using the 2.7m Harlan J Smith Telescope in order to determine the level of chemical homogeneity.

Observing Experience

2.7m Harlan J. Smith Telescope, McDonald Observatory (21 nights) — Ft Davis, TX

0.8m Telescope, McDonald Observatory (2 nights) — Ft Davis, TX

Service & Outreach

Poster Presenter, Astronomy on Tap Austin 100th Show	July 2026
Graduate Student Assembly Representative, The University of Texas at Austin	2025 - 2026
Student Panelist, UCLA AstroLive Summer Workshop	July 2022
Volunteer and Student Panelist, UCLA Physics and Astronomy Outreach Event	May 2022

Skills

Programming Languages: Python, SQL/ADQL, C/C++
Tools: iSpec, SAPHIRES, galpy
Software: BACCHUS, IRAF, SAOImageDS9
Other: Linux/Unix, Git, LaTeX